Equation Verification and Corrections

Verification pass: each PDF page re-examined against extracted knowledge files.

0.1 Chapter 2

0.1.1 Correction C2-1 Surface resistance formula scope *(p.16 of chap5, not chap2)*

No corrections to Chapter 2 equations. All verified against slide images: - Z0 general (slide 9) - Z0 microstrip (87/sqrt(_r+1.41)) » ln(5.98H/(0.8W+T)) (slides 9, 39) - Z0 stripline (60/sqrt(_r)) » ln(4H/(0.67(T+0.8W))) (slide 10) - v = c/sqrt(_r), PD = sqrt(_r)/c, TD = x_r/c (slide 11) - TD = LC, PD = (L_unit » C_unit) (slide 11) - Effective _e formula (slides 12, 40) - SPICE segments 10(x/v/T_r) (slide 14) - V_i = V_s » Z_0/(Z_0+Z_s) (slide 17) - = (Z_tZ_0)/(Z_t+Z_0) (slide 18) - Example 2.1 numbers: TD=898 ps, L_total=44.9 nH, C_total=17.9 pF, 4 segments (slides 1516) - Additional example: _e=2.84, v=1.78»10^8 m/s, TD=713 ps (slide 40)

0.2 Chapter 3

0.2.1 Correction C3-1 NEXT formula sign convention *(p.13)*

The FEXT formula from slide 13 is:

$$B_{\text{FEXT}} = -\frac{V_{\text{input}} \cdot X\sqrt{LC}}{2T_r}\left(\frac{L_m}{L} - \frac{C_m}{C}\right)$$

The negative sign means FEXT is negative if $L_m/L > C_m/C$ (inductive coupling dominates). This is correctly recorded.

0.2.2 Verification of all Chapter 3 equations:

- V_noise,Lm = L_m » dI/dt (slide 4)
- I_noise,Cm = C_m » dV/dt (slide 5)
- I_near = I(L_m) + I_near(C_m) (slide 11)
- I_far = I_far(C_m) I(L_m) (slide 11)
- NEXT amplitude A = V(input)/4 » (L_m/L + C_m/C) (slide 13)
- FEXT amplitude B = V(input)XLC/(2T_r) » (L_m/L C_m/C) (slide 13)
- T_r > 2TD: multiply NEXT by 2TD/T_r; FEXT unchanged (slide 14)
- L_odd = L_11 L_12 (slide 25)
- Z_odd = ((L_11L_12)/(C_11+C_m)) (slide 27)
- TD_odd = ((L_11L_12)(C_11+C_m)) (slide 27)
- L_even = L_11 + L_12 (slide 28)
- C_even = C_11 C_m (slide 29)
- Z_even = ((L_11+L_12)/(C_11C_m)) (slide 29)
- TD_even = ((L_11+L_12)(C_11C_m)) (slide 29)

0.3 Chapter 4

No equations extracted from text were verifiable via slide images for the estimation formulas on p.14 (complex multi-line expressions). These are flagged:

0.3.1 C4-1 NEXT/FEXT estimation equations *(p.14)*

The estimation equations for NEXT and FEXT peaks with front-end C_f were only partially captured by text extraction. The slide shows multi-line expressions involving Z_even, Z_odd, Z_0, and the decoupling capacitor time constant = R_t » 2C_f. Recommend re-reading chap4.pdf p.14 directly for exact forms before using these in calculations.

0.3.2 Verification of all other Chapter 4 equations:

- TD_even = ((L_s+L_m)C_s) (slide 10)
- TD_odd = ((L_sL_m)(C_s+2C_m)) (slide 10)
- TDc_odd = ((L_sL_m)(C_s+2C_m+2C_f/)) (slide 23)
- Results tables (FEXT, NEXT, eye diagram) (slides 2627, 3536, 43)

0.4 Chapter 5

0.4.1 Correction C5-1 Surface Resistance R_S definition *(p.16)*

Original (incorrect) in chap5.md: > $R_S = \sqrt{\rho\pi\mu}/W \cdot L$

Corrected (per slide 16): > $R_S = \sqrt{\rho\pi\mu}$ in units of ůs/square

The full expression is:

$$R_{\text{ac signal}} = \frac{L\sqrt{\rho\pi\mu F}}{W} = R_S\sqrt{F} \cdot \frac{L}{W} \quad \Omega/\text{m}$$

The surface resistance per square ($L = W$) is simply $R_S\sqrt{F}$. This has been corrected in `knowledge/chap5.md`.

0.4.2 Verification of all other Chapter 5 equations:

- = (2/) = (/F) (slide 6)
- R_ac microstrip (F)(1/W + 1/(6H)) /m (slide 12)
- Loss ratio formula (slide 12)
- R_ground formula with W/H term (slide 12)
- tan|_d| = 1/(2F) = / (slide 21)
- G = (/) » 2F » C_11 (slide 21)
- $C_{90^\circ\text{bend}}$ C_11 » w (slide 39)
- Fourier square wave expansion (slide 17)

0.5 Summary

Chapter	Corrections	VERIFY flags
Chap 1	0	0
Chap 2	0	0
Chap 3	0 (sign convention clarified)	0
Chap 4	0	1 (complex estimation eqs p.14)
Chap 5	1 (R_S definition corrected)	0